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Perceptual learning alters post-sensory processing in human decision-making

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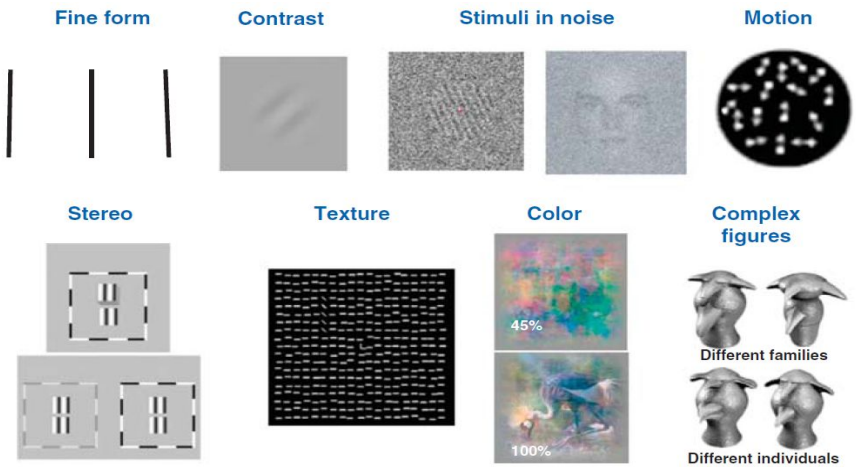
Introduction

What is the problem dealt and why is it important?

- The neural substrates of perceptual learning and how it affect decision-making is elusive
  - Perceptual learning - how experience can change the way we perceive sensory stimulus
- Examples of perceptual learning:

Radiologists can differentiate the healthiness of tissues from x-ray images (red circles: bone cancers)



Various modalities of perceptual learning

Introduction

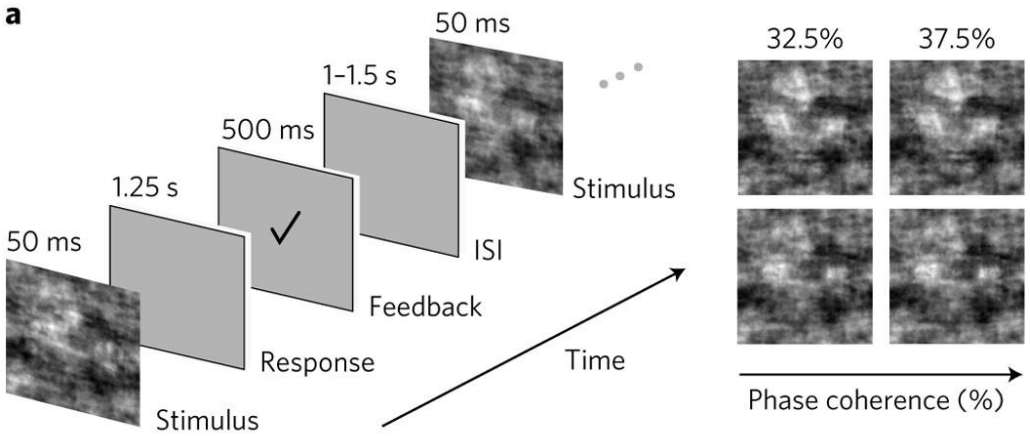
What is their unique approach?

- Trained participants on a visual categorization task over three days and used multivariate pattern analysis of the EEG to identify two temporally specific components
- Modelled the training improvements with a reinforcement learning mechanism using a reward prediction error signal

Results

1. Improvements in behavioral performance were accompanied primarily by late enhancements in decision-related evidence
2. The onset of the late component systematically moves earlier in time with training
3. The learning-induced behavioural improvements could be reliably explained in terms of a reinforcement learning mechanism

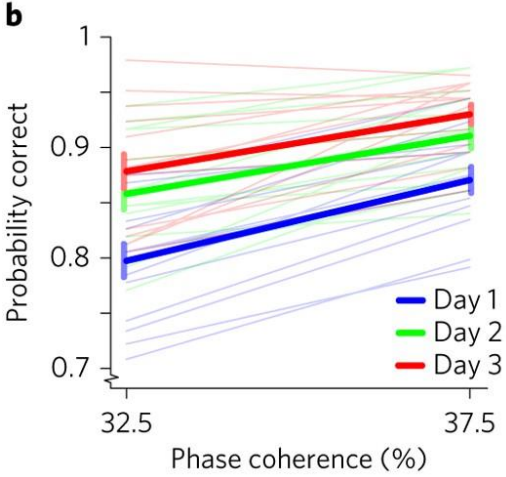
Experimental Design and Behaviour

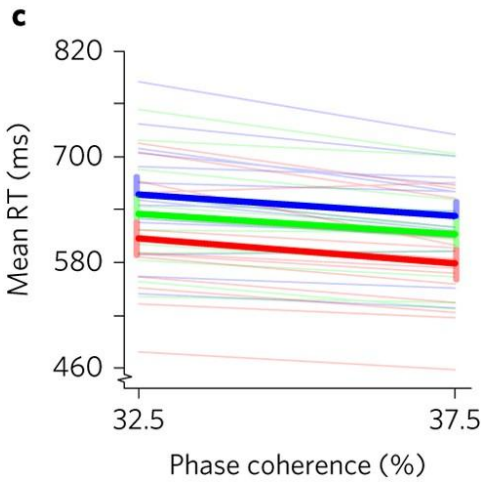


(a) Experimental paradigm

1. Binary categorization for a noisy image shown for 50 ms
2. Response (choice)
3. Visual feedback
4. Inter stimulus interval of 1 s

❖ Phase coherence is inversely related to difficulty



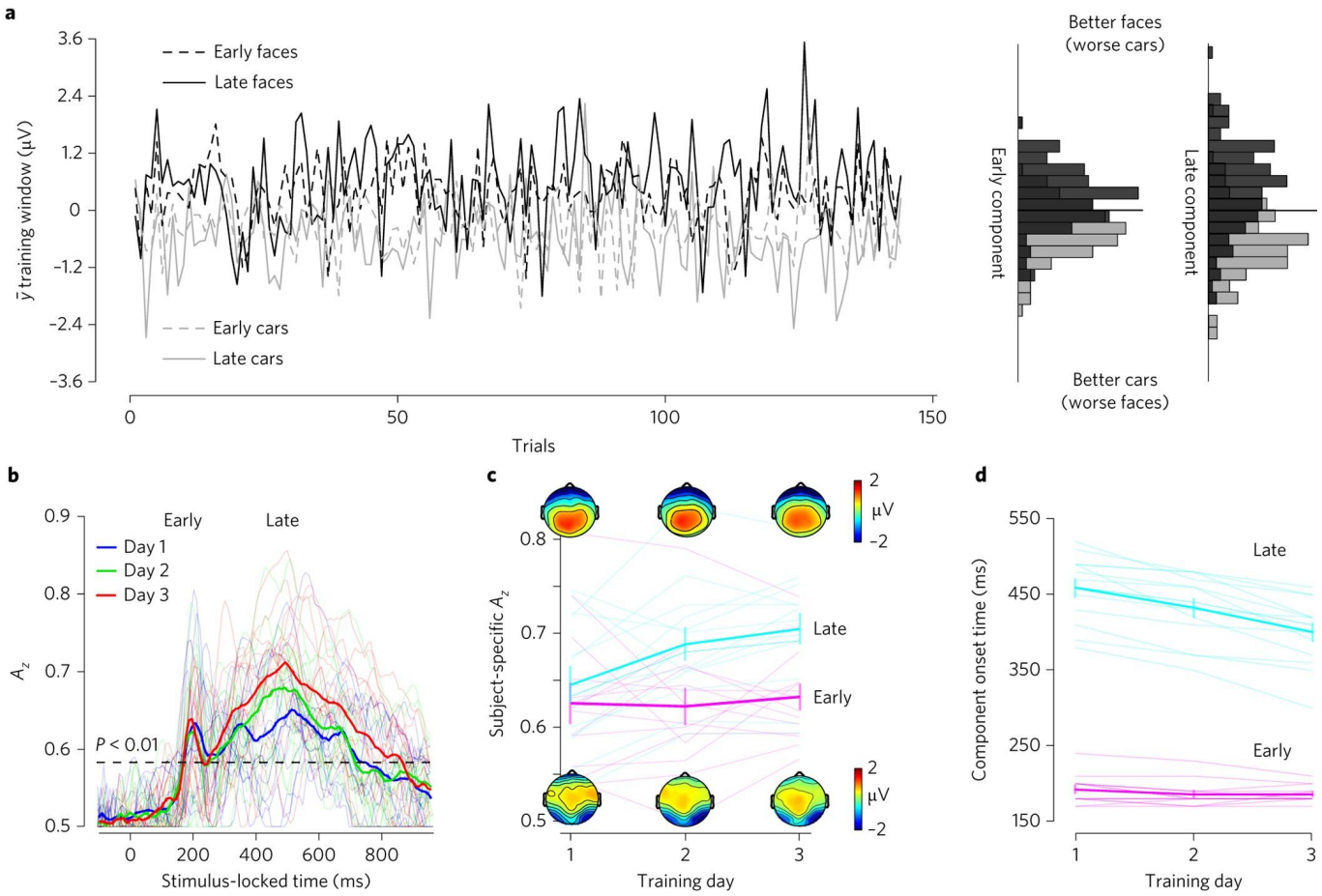


Results

(b & c) As expected,

- Training improved accuracy and reduced reaction time
- Difficulty is proportional to reaction time and inversely related to accuracy

Post-Sensory Effects of Perceptual Learning



y: discriminating comp. amplitude

Az: area under the ROC curve

Results

Analysis w.r.t two temporal comp.s

- ❖ Late Comp.: around 187 ms
- ❖ Early Comp.: around 431 ms

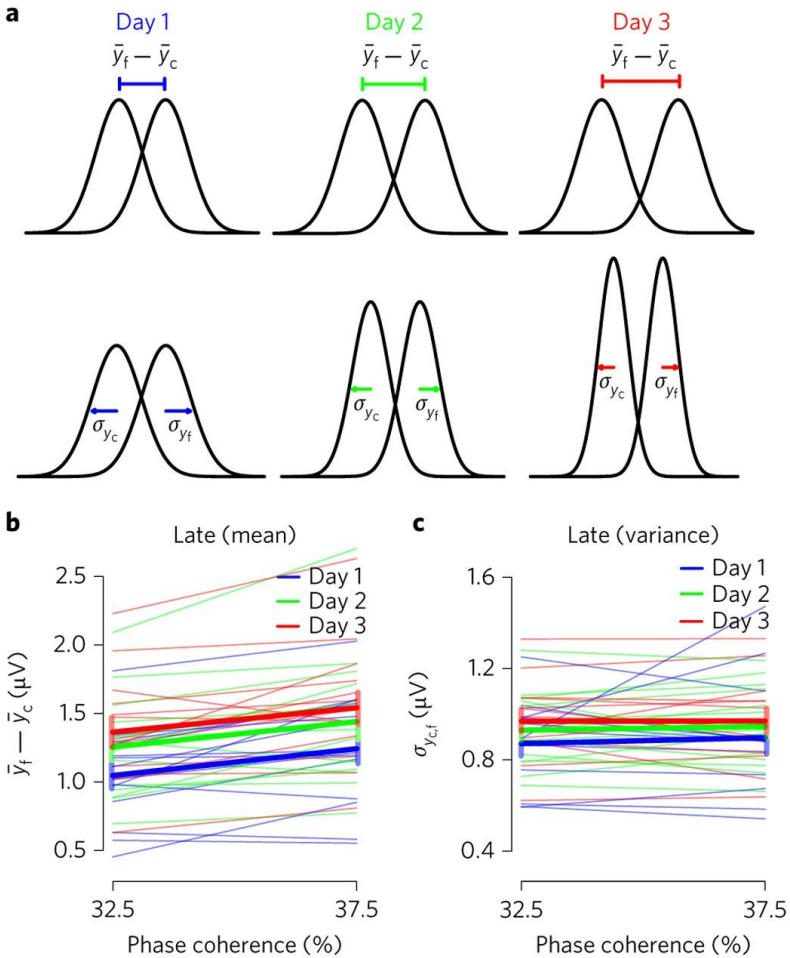
(c) Across the training, the discrimination performance of

- Late Comp. ↑
- Early Comp. didn't change
- ➔ Late Comp. is encoding the decision evidence

(d) The peak times of

- Late Comp. ↓
- Early Comp. didn't change
- ➔ Late Comp. is lined to the process of learning

Enhanced Readout of Post-Sensory Decision Evidence



Single-trial variability in the component amplitudes of Late Comp.

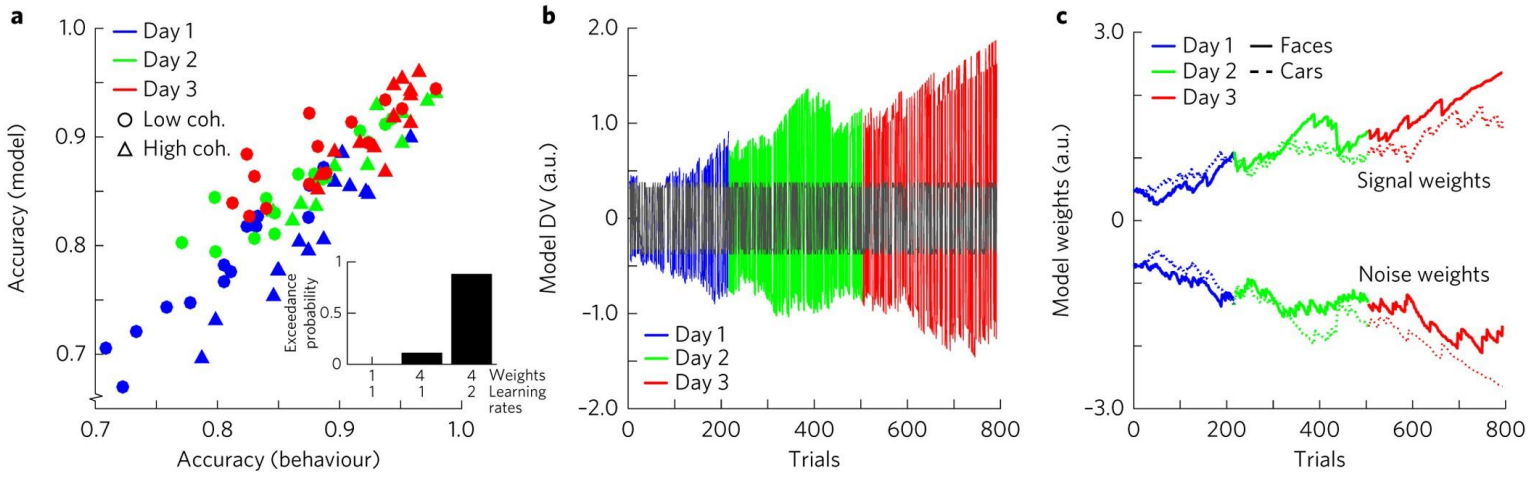
- ❖ yf-yc: distance between the mean face and car component amplitudes
- ❖ σ: trial-by-trial variability

Results

- Effect of the sensory evidence amount on the means, but not variance
- Effect of training day on the means, but not variance

➔ Improvements in the Late Comp. discrimination performance over training days are the result of gain modulation of the comp. amplitudes, rather than a reduction in the trial-to-trial variability

Reinforcement Learning Model for Perceptual Choices



Modelled the participant's choices using a reinforcement learning model

- Decision variable (DV): strength of the presented stimulus on a given trial (stimulus sensory evidence)
- Positive DV → higher likelihood of a face / Negative DV → higher likelihood of a car
- Weights are updated by means of a prediction error signal (discrepancy between the expected and actual value of the decision

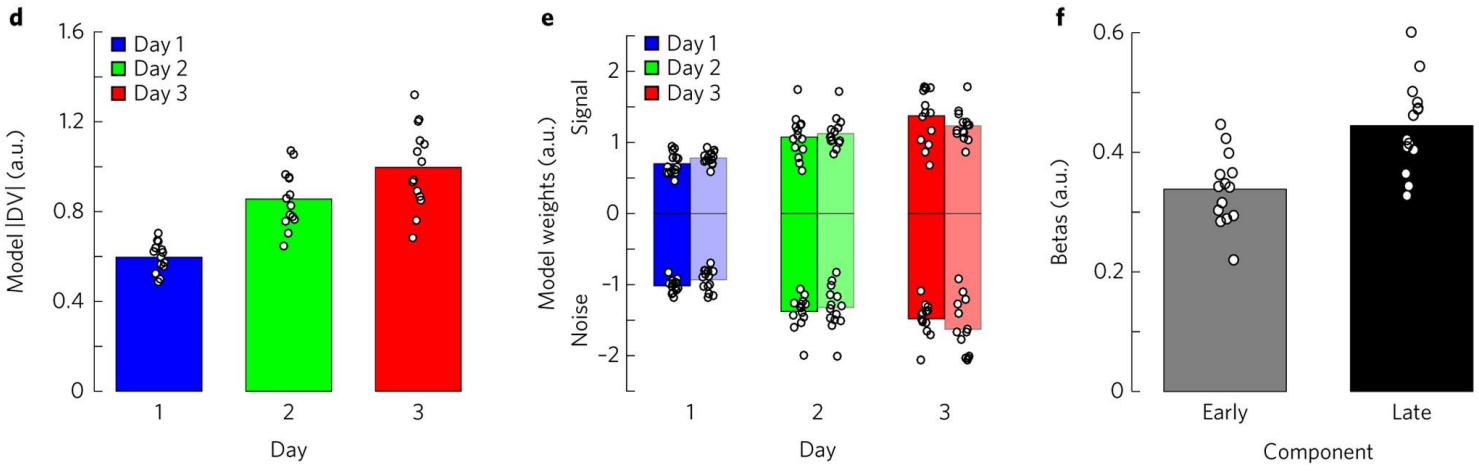
Results

(a) High correspondence between the accuracy of the model and actual behaviour

(b) Subject-wise gradual build-up in the trial-by-trial estimates of the DVs

(c) Increase in the signal weights mirrored by a decrease in the noise weight

Reinforcement Learning Model for Perceptual Choices



Modelled the participant's choices using a reinforcement learning model

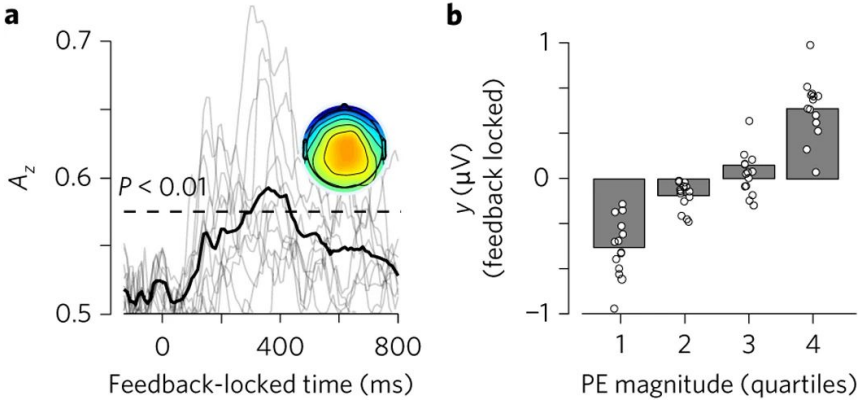
- Decision variable (DV): strength of the presented stimulus on a given trial (stimulus sensory evidence)
- Positive values → higher likelihood of a face / Negative values → higher likelihood of a car
- Weights are updated by means of a prediction error signal (discrepancy between the expected and actual value of the decision

Results

(d & e) Significant effect of learning: subject-wise mean DVs and aggregated perceptual weights

(f) Late Comp. was both a reliable and better predictor than the Early Comp.

Electrophysiological Correlates of Prediction Error



(a) Separated trials into four bins based on the model-predicted magnitudes of the prediction error (PE) signal and ran single-trial discriminant analysis on feedback-locked EEG data between the very low and very high PE trial groups

- Result: EEG component peaking on average at 354 ms post-feedback

➔ Consistent with work on feedback-related processing in human brain using a probabilistic reversal learning task

(b) Computed discriminator amplitudes (y) for trials with intermediate magnitude levels and applied the spatial filter with the highest discrimination performance for the extreme PE magnitude levels

- Result: Mean discriminator output for each quartile increased as a function of the mode's PE magnitude

➔ Link between the model's PE estimates and feedback-related EEG comp.

Conclusion

1. Improvements in behavioral performance were accompanied primarily by late enhancements in decision-related evidence

➔ Strengthening of the connections between early sensory encoding and downstream decision-related processing are driving perceptual learning in our task

2. The onset of the late component systematically moves earlier in time with training

➔ Learning effects on the late component lead to changes in perceptual sensitivity

3. The learning-induced behavioural improvements could be reliably explained in terms of a reinforcement learning mechanism

➔ Perceptual learning involves an enhanced readout of sensory information during decision-making, probably through a process similar to reinforcement learning on a domain general learning mechanism.

References

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